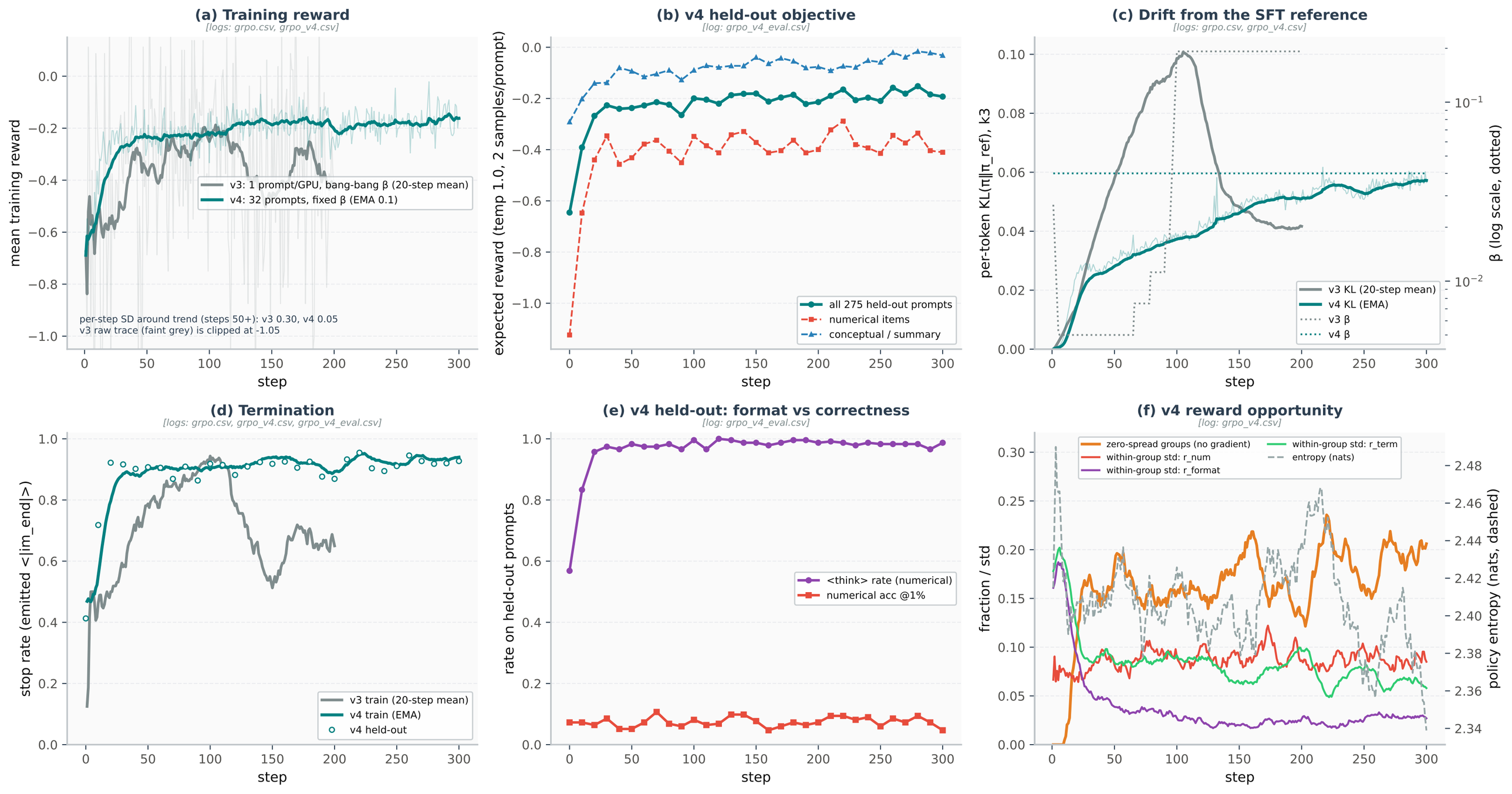

(a) Training reward
[logs: grpo.csv, grpo_v4.csv]
mean training reward
step
v3: 1 prompt/GPU, bang-bang β (20-step mean)
v4: 32 prompts, fixed β (EMA 0.1)
per-step SD around trend (steps 50+): v3 0.30, v4 0.05
v3 raw trace (faint grey) is clipped at -1.05
(b) v4 held-out objective
[log: grpo_v4_eval.csv]
expected reward (temp 1.0, 2 samples/prompt)
step
all 275 held-out prompts
numerical items
conceptual / summary
(c) Drift from the SFT reference
[logs: grpo.csv, grpo_v4.csv]
per-token KL(π‖π_ref), k3
β (log scale, dotted)
step
v3 KL (20-step mean)
v4 KL (EMA)
v3 β
v4 β
(d) Termination
[logs: grpo.csv, grpo_v4.csv, grpo_v4_eval.csv]
stop rate (emitted <|im_end|>)
step
v3 train (20-step mean)
v4 train (EMA)
v4 held-out
(e) v4 held-out: format vs correctness
[log: grpo_v4_eval.csv]
rate on held-out prompts
step
<think> rate (numerical)
numerical acc @1%
(f) v4 reward opportunity
[log: grpo_v4.csv]
fraction / std
policy entropy (nats, dashed)
step
zero-spread groups (no gradient)
within-group std: r_num
within-group std: r_format
within-group std: r_term
entropy (nats)